

Between War and Peace: The Cultural Symbolism and Military Applications of Pigeons Throughout History

Introduction: The Paradox of the Pigeon

The humble pigeon, a member of the widespread *Columbidae* family, presents a remarkable paradox in human history. Simultaneously revered as an iconic symbol of peace, love, and spiritual purity, and utilized as a crucial instrument of warfare across millennia, this species embodies a profound duality. This report undertakes an interdisciplinary analysis to unravel this paradox, examining the pigeon's multifaceted roles through the lenses of Science and Technology Studies (STS), cultural and symbolic analysis, and military history. The aim is to move beyond anecdotal accounts to a rigorous, theoretically informed examination of how a single animal can occupy such contradictory positions within human societies.

The significance of this investigation is manifold. It provides a comprehensive historical and cultural analysis of pigeons, re-evaluates the complex dynamics of human-animal military partnerships, and offers insights into the processes of military innovation and technological adoption. Furthermore, this study develops a framework for understanding how the same species can embody conflicting cultural meanings, bridging disparate academic fields to illuminate an understudied aspect of human-animal relations in both conflict and harmony.

1. The Biological and Behavioral Foundations of Pigeon Utility

The enduring utility of pigeons in military applications throughout history stems from a unique confluence of biological and behavioral attributes. These inherent capabilities were not merely exploited but actively refined and leveraged through human intervention, transforming the bird into a sophisticated bio-technological asset.

Homing Abilities and Navigational Capabilities

Pigeons possess an extraordinary innate ability to navigate vast distances and return to their home loft. This homing instinct, meticulously cultivated through selective breeding, enabled their use as messengers as far back as ancient Egypt, Rome, and Persia, a practice that continued for millennia.¹ Modern scientific inquiry into this remarkable capability suggests a multi-modal navigational system. Theories propose that pigeons employ a "compass mechanism," likely relying on the sun's position, and a "map mechanism" that may integrate Earth's magnetic fields (magnetoreception), low-frequency infrasound (creating a "sound map"), and recognition of familiar visual landmarks.² For instance, research indicates that while pigeons can utilize familiar landmarks for navigation, the specific mechanisms differ, with some birds relying on direct "pilotage" (spatial relationships between landmarks) and others on "site-specific compass orientation" (recalling the compass direction from a familiar location).⁵

The multi-modal nature of pigeon navigation provided a robust and resilient communication system, a characteristic that proved strategically invaluable. If one navigational cue was compromised—for example, a cloudy day obscuring the sun compass—other mechanisms could compensate. This inherent biological redundancy rendered pigeons less susceptible to simple countermeasures or environmental disruptions that could cripple early mechanical or electronic communication systems. The fact that the precise mechanisms of pigeon navigation remain not fully understood² meant that their communication method was inherently resistant to enemy interception or jamming in ways that electronic signals, with their known vulnerabilities, were not. This "black box" quality of their biological navigation was not merely a scientific curiosity; it represented a strategic advantage, highlighting how complex natural systems, even when not entirely comprehended, can offer unique operational security benefits that engineered systems often struggle to replicate. This observation underscores a broader pattern in military innovation, where "low-tech" or "natural" solutions occasionally demonstrate counter-intuitive utility precisely because their underlying mechanisms are opaque or difficult to counter by conventional means.

Visual Acuity and Pattern Recognition

Pigeons exhibit superior visual acuity and a broad field of vision compared to humans, enabling them to efficiently scan large areas and discern specific patterns.⁸ Their capacity for sustained visual vigilance over extended periods was particularly noteworthy.⁸ This keen vision and pattern recognition ability were central to proposed military applications, most notably B.F. Skinner's Project Pigeon, where birds were trained to peck at target images displayed on a screen within a missile's nose cone.¹⁰

The success of projects like the U.S. Coast Guard and Navy's "Project Sea Hunt" further illustrates the practical advantage of avian vision. In this program, pigeons housed beneath rescue helicopters were trained to identify red, yellow, or orange objects in the ocean, such as life jackets. During experiments, these avian observers spotted test targets on initial passes 90% of the time, a rate significantly higher than the less than 50% achieved by human aviators in the same tests.⁸ This remarkable performance underscores a specific biological superiority: pigeons' sustained visual vigilance compensated for the rapid onset of human eye fatigue, providing a robust, albeit unconventional, solution where human operators were limited. This phenomenon reveals a historical phase in technological development where organic systems were not merely substitutes but, in certain demanding tasks, demonstrably superior alternatives to nascent mechanical or electronic control systems. It highlights a fascinating moment of human-animal-machine hybridity driven by pragmatic military needs, where biological "automation" offered unique advantages in precision and endurance.

Flight Speed, Endurance, and Maneuverability

Homing pigeons exhibit impressive flight characteristics crucial for rapid and reliable communication. They are capable of traveling 100 to 600 miles in a single flight, maintaining average speeds of 35-60 mph, with bursts reaching up to 90 mph over shorter distances.¹⁴ This combination of speed and endurance allowed for swift message delivery, often surpassing the pace of human messengers.⁷

Beyond sheer speed, their maneuverability and ability to fly at high altitudes made them challenging targets for enemy fire and shrapnel.¹⁷ Their resilience was further demonstrated by the practice of dropping them from aircraft traveling at speeds up to

375 mph, often with minimal protection such as an ordinary paper grocery bag, from which they would emerge and spiral down before flying home.¹⁴ Pigeons also proved highly adaptable to diverse and chaotic battlefield conditions, operating effectively despite bombardments, dust, smoke, and adverse weather.¹ This superior performance in unpredictable environments, where early electronic communication methods like telegraph wires were easily cut or radios were unreliable and prone to jamming, underscores a critical period when biological systems offered greater resilience and reliability. The limitations of early mechanical communication directly necessitated reliance on pigeons, whose inherent biological attributes provided a more robust and effective alternative in those challenging operational contexts.

Susceptibility to Operant Conditioning and Trainability

Pigeons are highly adaptive and notably amenable to training, capable of achieving combat readiness in as little as eight weeks.⁸ Their high susceptibility to operant conditioning, a method famously developed and applied by B.F. Skinner, allowed for the precise shaping of their behavior through systematic reinforcement.¹⁰ Military training regimens for pigeons were sophisticated, involving a gradual increase in flight distances and the strategic use of powerful motivators such as hunger, jealousy, and even sex to ensure rapid and consistent returns to their lofts.¹⁴

These detailed training protocols and the application of psychological motivators demonstrate a sophisticated form of behavioral engineering applied to optimize animal performance for military objectives. This highlights that human-animal partnerships in warfare were not merely about exploiting innate abilities but involved active, scientifically informed manipulation of animal behavior to meet specific operational requirements. The remarkable tenacity of conditioned pigeons was evidenced in Skinner's experiments, where they continued to peck accurately even when subjected to simulated combat conditions, including loud noises, pressure changes simulating high altitudes, intense G-forces in centrifuges, and bright flashes mimicking shell bursts.¹¹ This psychological robustness further solidified their potential as reliable components within military systems.

Adaptability to Diverse Environments and Conditions

The adaptability of pigeons allowed for their deployment across a wide array of military environments and conditions. They were utilized in the confined spaces of trenches, on board naval vessels, and released from aircraft, demonstrating their versatility across land, sea, and air operations.¹ Their resilience extended to surviving extreme physical conditions, such as being dropped from high-speed aircraft with minimal protection.¹⁴ Their ability to operate reliably amidst heavy shell-fire, dust, smoke, and adverse weather conditions made them indispensable when more conventional communication methods—like fragile telegraph lines or easily intercepted human messengers—proved ineffective.¹ This inherent hardiness ensured their operational reliability in the unpredictable and chaotic environments of warfare.

Table 1: Key Biological and Behavioral Features of Pigeons and Their Military Relevance

Feature	Description	Military Relevance
Homing Ability	Innate directional sense; ability to return to a specific loft from long distances (up to 1100 miles).	Reliable, jam-proof communication, especially when other methods failed; low-visibility intelligence transport.
Visual Acuity & Vigilance	Superior field of vision, high contrast sensitivity, ability to maintain focus on visual tasks for hours.	Aerial reconnaissance, search and rescue (e.g., Project Sea Hunt), target recognition for guidance systems (e.g., Project Pigeon).
Trainability (Operant Conditioning)	Highly adaptive and responsive to reinforcement; capable of complex learned behaviors.	Message carrying, target pecking for missile guidance, specific task performance under stress.
Flight Speed & Endurance	Average 35-60 mph, bursts up to 90 mph; distances up to 600 miles.	Rapid message delivery, evading enemy fire, covering vast distances quickly.
Physical Resilience & Adaptability	Can withstand G-forces, pressure changes, loud noises, and adverse weather; can be dropped from high altitudes.	Survivability in combat zones, operational reliability in diverse environments, deployment flexibility.

This table systematically outlines the specific biological and behavioral attributes of pigeons and their direct military applications. It serves to highlight that pigeons were not merely animals incidentally used in war, but rather bio-technological assets whose natural endowments were systematically leveraged and optimized for strategic purposes. This structured presentation provides a foundational understanding of *why* pigeons were chosen for various roles, including radical proposals like Skinner's missile guidance system.

2. Project Pigeon/Orcon: A Case Study in Animal-Machine Integration

B.F. Skinner's Project Pigeon, later known as Project Orcon (Organic Control), represents one of the most unconventional yet scientifically rigorous attempts at animal-machine integration in military history. This ambitious program sought to leverage the unique cognitive capabilities of pigeons to guide missiles, a concept born from the critical need for bombing accuracy during World War II before the advent of reliable radar and electronic guidance systems.¹²

Skinner's Behavioral Approach and Technical Implementation

Skinner, a leading figure in behaviorism, proposed a system where pigeons, trained through operant conditioning, would serve as the guidance mechanism for missiles. The technical design was ingenious: a missile nose cone was fitted with three separate compartments, each containing a pigeon. An image of the target, such as an enemy ship, would be projected onto a screen within each compartment. The pigeons were meticulously trained to peck at this target image, and their pecking, detected by a metal conductor attached to their beak, would generate signals. These signals would then be transmitted to the missile's guidance system, controlling its tail fins to keep it precisely on course.¹⁰ The pigeons were rewarded with food for maintaining the target in the center of their screen, reinforcing the desired behavior.¹⁰

Skinner's preliminary tests demonstrated remarkable accuracy, with the pigeons achieving "nearly perfect" performance.¹¹ To assess their psychological fitness for combat, Skinner subjected the birds to rigorous simulated battlefield conditions.

These tests included exposure to loud noises (pistol shots), pressure changes simulating altitudes of 10,000 feet, massive G-forces in a centrifuge, and bright flashes designed to mimic shell bursts. Despite these extreme stressors, the pigeons consistently maintained their pecking behavior with high accuracy.¹¹ Some pigeons were observed pecking over 10,000 times in a 45-minute period²⁰, underscoring their sustained focus and trainability. The long-term retention of these conditioned behaviors was also impressive, with some pigeons retaining their skills "sharp enough for battle" even "one, two, four, and even six years later".¹³

Military Reception and Evaluation of the Pigeon-Guided Missile Concept

Despite the demonstrable technical success and the pigeons' remarkable performance under simulated combat conditions, military authorities largely met Project Pigeon with skepticism. The concept was widely regarded as "too eccentric and impractical".²⁵ Although the National Defense Research Committee initially provided \$25,000 in funding for the research, the project was ultimately canceled in October 1944. The official rationale cited was that "further prosecution of this project would seriously delay others which in the minds of the Division have more immediate promise of combat application".²⁵

The project was briefly revived by the U.S. Navy in 1948 under the new name "Project Orcon." However, it faced a similar fate, being canceled definitively in 1953. The primary reason for this final discontinuation was the proven reliability and increasing sophistication of electronic guidance systems, which were deemed more practical and necessary for the evolving landscape of missile technology.¹¹ Beyond the technical advancements, there was a subtle, yet significant, underlying factor: administrators reportedly struggled to reconcile with the ethical implications of pigeons dying in kamikaze missions for the cause.²⁵

The rejection of Project Pigeon, despite its technical viability, illustrates the powerful influence of what might be termed the "giggle factor" and the social construction of technological legitimacy. Skinner himself famously lamented, "Our problem... was no one would take us seriously".¹¹ This suggests that the project's failure was not solely due to technical limitations but also stemmed from a cultural bias against animal-based technology in favor of what was perceived as "less eccentric" or more "advanced" mechanical solutions. This highlights a tension between scientific efficacy and military cultural acceptance, where the aesthetic and perceived sophistication of

technology can override pragmatic technical success. The subsequent public presentation of Skinner's pigeons, after the research was declassified, focused on their ability to play ping-pong and piano, deliberately omitting their proposed wartime duties.¹¹ This deliberate re-framing underscores a strategic management of public perception regarding scientific endeavors, particularly those involving sensitive military applications or potentially controversial animal research, shaping the meaning and acceptance of the technology.

Comparison with Norbert Wiener's Cybernetic Approach to Anti-Aircraft Systems

The mid-20th century, particularly the World War II era, witnessed parallel intellectual developments in the fields of control and information theory, exemplified by B.F. Skinner's behaviorism and Norbert Wiener's cybernetics. Both theoretical frameworks shared a foundational interest in feedback loops and control systems.²⁸ Skinner's operant conditioning, predicated on the principle of reinforcement, bears an analogy to Wiener's concept of information loops and feedback mechanisms, both aiming to shape or predict outcomes through recursive processes.²⁸

Norbert Wiener, widely recognized as the originator of cybernetics, focused on the science of communication and control in both living organisms and machines.²⁹ His work was heavily influenced by his World War II experiments with anti-aircraft systems, which sought to predict the trajectory of enemy planes by interpreting radar images and using feedback loops to adjust defensive fire.²⁹ Wiener coined the term "cybernetics" from the Greek word for "steersman," emphasizing the concept of self-regulation and control through information flow.³⁰

The fundamental difference between Skinner's and Wiener's approaches, despite their shared interest in control, lay in their chosen mediums and conceptualization of intelligence. Skinner pursued an embodied intelligence model, where the animal's trained biological "computation" directly interfaced with the mechanical system, using the pigeon's natural sensory and motor capabilities as a living component of the missile's guidance system.¹⁰ Wiener, conversely, pioneered an abstracted intelligence model. His cybernetics focused on the theoretical principles of information processing, prediction, and noise reduction, which could then be applied to purely mechanical and later digital computation, such as predicting enemy aircraft movements to inform anti-aircraft gun adjustments.²⁹

This comparison highlights a fundamental divergence in the conceptualization of control and intelligence during a pivotal period of technological development. Skinner's "organic control" was a direct, albeit controversial, attempt to leverage biological processing power for a specific task. Wiener's cybernetics, however, laid the theoretical groundwork for purely mechanical and later digital computation, ultimately proving more scalable and socially acceptable for complex military applications. The success of Wiener's cybernetics in developing reliable electronic guidance systems directly contributed to the obsolescence of animal-based systems like Project Pigeon.²⁵ This illustrates a clear technological displacement, where a more abstract, mathematical, and ultimately scalable control theory superseded a more concrete, biological, and less scalable one. This shift also occurred within a broader sociopolitical context following World War I, where there was a perceived "crisis of authority" and a desire to establish "new and lasting forms of social control".²⁸ Both behaviorism and cybernetics, in their distinct ways, offered frameworks for managing complex systems—whether human or machine—reflecting a societal yearning for order and predictability in a chaotic world.

3. Pigeons in Warfare: Beyond Message Carriers

While their role as message carriers is widely recognized, pigeons have been employed in a surprising array of military applications throughout history, offering unique advantages that mechanical alternatives of their time could not replicate.

Message Carrying (Primary Role)

For millennia, pigeons served as the backbone of military communication. From Cyrus the Great of Persia in the 6th century BC and Julius Caesar in Ancient Rome, to their extensive use in the Franco-Prussian War (1870–71), and both World Wars, pigeons provided a reliable means of transmitting vital information.¹ They were particularly indispensable in situations where telegraph lines were cut, early radio technology was unreliable or inaccessible, or human messengers faced insurmountable risks of capture or death.¹ During World War I, pigeons achieved a remarkable 95% success rate in delivering messages, a testament to their reliability in chaotic and dangerous environments.¹ Their ability to cut through the "silence" of battle with wings, without

reliance on fragile wires or limited-range radios, made them invaluable.¹⁵

Aerial Reconnaissance and Photography

Beyond simple message delivery, pigeons were adapted for more sophisticated intelligence gathering. During World War I, some pigeons were equipped with small, unmanned cameras designed to automatically take surveillance photographs as they flew over enemy positions.⁴ This early form of aerial reconnaissance provided valuable visual intelligence from areas difficult or dangerous for human observation. Pilots also utilized pigeons, launching them mid-air from their aircraft to rapidly transmit reconnaissance information back to commanding officers on the ground.⁴ This allowed for quick updates on battlefield conditions or enemy movements. The potential of pigeons for aerial photography continued to be explored even after the World Wars, with the CIA later patenting specialized photography equipment specifically for these birds to capture aerial footage.³⁷

Search and Rescue Operations

Pigeons played a critical role in search and rescue missions, particularly during World War II. Numerous pigeons were credited with saving the lives of downed Allied aircrew, with 32 receiving the prestigious Dickin Medal for their heroic actions in delivering distress messages.³⁸ A notable example is the U.S. Coast Guard and Navy's "Project Sea Hunt," a pigeon-powered sensor system developed for ocean search and rescue. In this project, trained pigeons housed in a "command center" box beneath rescue helicopters were taught to identify red, yellow, or orange objects—colors commonly used for life jackets and emergency gear—in the vast expanse of the ocean. When a pigeon spotted such an object, it would peck a switch, alerting the human crew. Experiments demonstrated that these avian observers were significantly more vigilant than human crews, spotting test targets on initial passes 90% of the time, compared to less than 50% for human aviators.⁸ This highlights a specific biological advantage: pigeons' superior visual vigilance and tireless attention compensated for the rapid onset of human eye fatigue, making them highly effective for complex visual search tasks.

Psychological Warfare and Propaganda Delivery

Pigeons also served, albeit indirectly, in psychological warfare and propaganda efforts. British intelligence, for instance, conducted "Source Columba," an unconventional operation during World War II. This involved air-dropping homing pigeons behind enemy lines in German-occupied France and the Netherlands, accompanied by thin paper, special pencils, and instructions for local citizens to write down observations, such as troop movements, for Allied intelligence.⁴ A recent copy of a London newspaper was often included as "proof of good faith".⁴¹ This initiative not only gathered vital intelligence but also served as a form of psychological operations by fostering resistance and a sense of connection with Allied forces among the occupied populations. German counterintelligence recognized this threat and attempted to counteract "Source Columba" by dropping their own pigeons with misleading instructions, an effort that the Resistance countered by encouraging people to "smoke the cigarettes and eat the pigeons".⁴¹

Pigeons functioned as a primitive, yet remarkably effective, "distributed sensor network" for intelligence gathering. Their inherent biological characteristics, such as being "undetectable" ⁴³, "silent and untraceable" ¹⁵, having "zero electromagnetic emissions or obvious detectability to radar" ⁴⁴, and their inability to "talk under interrogation" ⁴⁴, made them ideal for clandestine missions. They could blend into the local avian population, providing a unique form of stealth that human or electronic messengers could not replicate.⁴⁴ The success of these operations, yielding useful intelligence in over 50% of received messages ⁴¹, demonstrates the strategic value of leveraging animal characteristics for covert operations, particularly in environments where human networks were compromised or electronic surveillance was prevalent. The act of a pigeon arriving with a message could also function as a "trust agent," fostering a sense of connection and reliability with resistance movements, an effect difficult to achieve with purely mechanical means.

Deception Operations and Counterintelligence

The strategic value of pigeons in warfare led to the development of countermeasures and deception tactics by opposing forces. As early as the Franco-Prussian War

(1870-71), the besieging Prussian Army employed hawks to hunt carrier pigeons used by besieged Parisians.³³ During World War I, German forces deployed sharpshooters and even falcons to intercept Allied pigeons, recognizing their critical role in communication.²² In response, French forces resorted to camouflaging their pigeons by dyeing them black to resemble crows, an early example of biological counter-countermeasures.²²

Beyond direct interception, pigeons were also integrated into deception operations. German counterintelligence during World War II, for example, dropped their own pigeons, disguised as British birds, with instructions designed to mislead Allied intelligence and identify local patriots.⁴¹ The capture of enemy pigeons, such as the German war pigeon "Kaiser" by American troops in WWI, also served a counterintelligence purpose. "Kaiser" was not only held as a prisoner of war but was integrated into the U.S. Army's breeding program, siring over 75 birds for use in WWII.¹⁹ This practice demonstrates not just tactical capture but a strategic integration of enemy assets into one's own military infrastructure, highlighting the efficiency and value placed on leveraging existing trained stock.

Modern Adaptations of Pigeon Capabilities in Drone and Autonomous Systems Design

While pigeons have largely been superseded by electronic communication and drone technology in modern warfare, some of their inherent advantages remain conceptually relevant and even inspire contemporary design. Pigeons offer a low-visibility option for relaying information, possessing "zero electromagnetic emissions or obvious detectability to radar".⁴⁴ This makes them inherently difficult to detect and track, a critical advantage in an era of advanced electronic warfare.⁴⁴ Their ability to carry small payloads, such as microSD memory cards containing gigabytes of video or imagery, represents a "low-technology option" for clandestine data transport that is resistant to electronic interception or jamming.⁴⁴

Furthermore, pigeons possess a "better power to weight ratio than drones" and are "self fueling," capable of evasive maneuvers that make them difficult targets.⁴⁴ These biological characteristics highlight that even in the modern era, certain natural advantages persist for niche military applications where stealth, autonomy, and resilience against electronic countermeasures are paramount. The efficiency of pigeon-based communication even inspired early research that contributed to the

development of computer data transmission systems.⁷ In a notable demonstration in 2009, a pigeon carrying data on a microSD card outperformed a leading South African internet service provider, delivering the data faster than 4% of the same data could be received over the internet.⁷ This suggests that biological systems can, in specific contexts, still serve as benchmarks for or even outperform advanced mechanical systems, challenging a purely linear view of technological progress.

Table 2: Comparative Advantages of Pigeons in Military Communication (Pre-Modern Electronics)

Advantage	Pigeon Characteristic	Comparison to Alternatives (e.g., Human Runner, Telegraph, Early Radio)
Reliability	Innate homing instinct; high success rate (95%); unaffected by terrain/weather.	Human runners vulnerable to capture/death; telegraph lines easily cut; early radio unreliable, bulky, easily intercepted/jammed.
Stealth/Undetectability	Small size, high altitude flight; no electromagnetic emissions; can blend with wild birds.	Human runners/motorcycles visible; radio signals detectable/locatable; no "signature" for radar.
Jam-Proof	Biological navigation, not reliant on electronic signals.	Radios susceptible to jamming; telegraphs to cutting.
Speed (over certain distances/conditions)	Average 35-60 mph, faster than ground messengers; can travel up to 600 miles.	Faster than foot/horse messengers, especially over rugged terrain; competitive with early vehicles/aircraft in certain scenarios.
Cost & Maintenance	Relatively low cost to acquire and maintain; self-sustaining (can find food/water).	High cost of electronic equipment; complex maintenance and power requirements for early electronics.
Psychological Resilience	Can withstand combat stress (noise, G-forces, pressure); maintain vigilance for hours.	Human fatigue; psychological impact on human messengers; early electronics prone to mechanical failure.

This comparative table systematically illustrates the specific niches where pigeons offered distinct and often irreplaceable advantages over the nascent mechanical and electronic communication technologies of their time. It demonstrates that their utility was not merely a fallback option but a strategic choice based on their unique attributes, such as being jam-proof, stealthy, and resilient in chaotic environments. This perspective reinforces the understanding that non-human actors, like pigeons, were integrated into socio-technical systems not out of primitivism, but due to their specific, superior capabilities in particular operational contexts.

4. Cultural Symbolism: The Pigeon-Dove Duality

The cultural symbolism of pigeons, particularly the semantic distinction between "pigeon" and "dove," reveals a profound duality in human perception and cultural construction. This section analyzes how these birds have been imbued with contradictory meanings across religious, political, and popular cultural contexts.

Religious Symbolism of Doves Across Different Faith Traditions

The dove has a long and rich tradition as a sacred symbol across diverse faith traditions, often representing purity, peace, and divine presence. In **Judaism and Christianity**, the dove is an iconic symbol of peace and the Holy Spirit.⁴⁵ The most enduring image is from the biblical story of Noah's Ark, where a dove returns with an olive branch, signifying the end of the flood and the restoration of God's peace on Earth.⁴⁵ In the New Testament, the Holy Spirit descends "like a dove" during Jesus' baptism, further cementing its spiritual significance.⁴⁷ Early Christians incorporated the dove and olive branch into funerary art to symbolize the peace of the soul.⁴⁹

In **Islam**, pigeons are regarded as pure animals, with a tradition that they saved Prophet Muhammad by nesting outside his hiding cave, thus deceiving his pursuers.⁴⁵
Ancient

Mesopotamian cultures prominently featured doves as symbols of Inanna-Ishtar, a goddess associated with love, sexuality, and war, appearing on cultic objects as early as the third millennium BC.⁴⁹ This association with a deity embodying both creation

and destruction suggests a deeper, more primal understanding of the bird's duality than the modern peace/war dichotomy. In ancient

Greek mythology, pigeons and doves were symbols of Aphrodite, the goddess of fertility, beauty, and love.⁴⁵

Native American cultures also held pigeons in high regard, using them as both a food source and in religious contexts; for example, the Blackfoot people considered the Mourning Dove a symbol of protection.⁴⁵ In

Chinese culture, pigeon symbolism dates back to at least the pre-Qin Dynasty, representing filial piety from the Han Dynasty onwards, and symbolizing spring as a transformation from a hawk.⁴⁵

The transformation of the dove's symbolism, from an ancient Mesopotamian association with a goddess of "love, sexuality, and war" (Inanna-Ishtar) to a Christian symbol of the "Holy Spirit" and "peace," reveals a profound cultural re-appropriation and re-contextualization of symbolism across civilizations and religions.⁴⁸ This demonstrates how cultural meanings are fluid, capable of being inverted or adapted over millennia to fit evolving theological or philosophical frameworks. Furthermore, the use of doves for sacrificial offerings in ancient Israel ⁴⁸ stands in stark contrast to its later symbolism of peace and purity, highlighting the complex and sometimes contradictory ways in which humans have interacted with and assigned meaning to animals, reflecting different facets of human belief systems, from appeasement to aspiration.

Political Use of Dove Imagery in Peace Movements

The dove has become a potent and widely recognized symbol of peace and pacifism in political discourse. Its imagery frequently appears in political cartoons, on banners at peace rallies, and in pacifist literature.⁴⁹ The most significant popularization of the dove as a secular peace symbol occurred after World War II, largely driven by Pablo Picasso's iconic lithograph,

La Colombe (The Dove), created in 1949.⁴⁹ This image was adopted as the emblem for the World Peace Council in Paris, and Picasso himself explicitly linked his art to the cause of peace, declaring, "I stand for life against death; I stand for peace against

war".⁴⁹

The dove symbol was subsequently used extensively in the post-war peace movement, including by the British Campaign for Nuclear Disarmament (CND) in the 1950s and 1960s.⁵⁰ This adoption by anti-war and counterculture activists cemented its status as a universal emblem of peace, hope, and social justice. The deliberate leveraging of this symbol by artistic and political movements demonstrates the immense power of such cultural elements to shape public discourse and attitudes towards peace and conflict. However, the symbol was not immune to ideological subversion; anti-communists, for instance, distributed posters depicting Picasso's peace dove "metamorphosing into a Soviet tank" ⁵⁰, illustrating how even a seemingly universal symbol can become a contested battleground in symbolic warfare.

Heroic Narratives of War Pigeons in Popular Culture and Media

Despite the dove's pervasive association with peace, the practical reality of pigeons' military service generated a distinct set of cultural narratives centered on their heroism. Stories of individual war pigeons became famous on the home front, with many surviving birds paraded alongside human veterans in victory celebrations.¹ Pigeons like Cher Ami, who saved the "Lost Battalion" in WWI by delivering a crucial message despite severe injuries ⁶, and G.I. Joe, who saved a village in Italy during WWII by calling off an impending air raid ³⁷, became national heroes. These birds received prestigious military honors, including the French Croix de Guerre and the British Dickin Medal.⁴

The creation and widespread awarding of the Dickin Medal, instituted in 1943 to honor "conspicuous gallantry or devotion to duty displayed by animals serving with the Armed Forces or Civil Defence units" ⁴⁰, signifies a formal institutional recognition of animals as legitimate, heroic participants in warfare. Often referred to as "the animals' Victoria Cross" ⁴⁰, this award transcends mere utilitarian use, acknowledging the agency and sacrifice of non-human actors. The act of attributing human-like qualities of bravery and devotion to animals served not only to recognize their vital practical contributions but also to bolster human morale, create relatable narratives of heroism for public consumption, and reinforce a sense of shared purpose across species in wartime. This highlights a complex interplay between pragmatic military logic and the cultural need for symbolic meaning and emotional connection, even with non-human

agents.

Transformation of Pigeons from Valued Military Assets to Urban "Flying Rats"

A striking aspect of the pigeon's duality is its dramatic transformation in popular perception from a valued military asset to an urban nuisance. Despite their celebrated past as "celebrated war heroes"⁴⁵, pigeons, particularly the common urban rock pigeon, have fallen out of favor in modern times. They are frequently portrayed negatively in media and colloquially described as "rats with wings".⁴⁵ This cultural degradation is largely driven by concerns over their potential to spread diseases and the accumulation of their droppings in urban environments.⁴⁵

This profound shift in perception illustrates a dynamic and often utilitarian "social contract" between humans and animals. When pigeons served a critical, irreplaceable military function, their presence was tolerated and even celebrated. Their stealth, jam-proof communication, and resilience were highly valued.¹⁵ However, as technological advancements such as radio and radar rendered their military role largely obsolete¹⁹, and as their populations concentrated in urban areas, their biological traits that were previously overlooked or managed (e.g., droppings, perceived disease risk) became salient "problems." This suggests that the value and symbolic meaning of animals are not inherent but are dynamically constructed based on their functional integration into human systems and the socio-economic costs or benefits they present. This phenomenon highlights a utilitarian dimension to human-animal relations, where cultural status is intrinsically linked to perceived utility and environmental context.

Cultural Representations of the Pigeon/Dove Duality in Art, Literature, and Propaganda

The duality of the pigeon/dove is deeply embedded in cultural representations across various artistic and linguistic forms. Pigeons and doves have been featured in art and literature for millennia, from ancient Egyptian paintings and Greek sculptures to the works of modern masters like Pablo Picasso.⁴⁵

A key element of this duality is the linguistic and cultural distinction between "pigeon" and "dove." Scientifically, both terms refer to birds within the same family, Columbidae, and are often interchangeable.⁵⁶ However, colloquially, "dove" (a word of Nordic origin) is typically used for smaller, often white, or more aesthetically pleasing birds, while "pigeon" (derived from French) commonly refers to the larger, often grey, urban bird.⁵⁸ This semantic divergence is not merely linguistic; it reflects a deliberate cultural preference and a desire to separate the positive, ancient symbolism from the more mundane or negative contemporary associations.

This linguistic split functions as a form of "semiotic engineering," a cultural process where language is employed to manage and compartmentalize contradictory meanings attributed to the same entity. By assigning the positive, sacred, and peaceful connotations to the "dove"—often depicted as white, associated with purity and ancient myths like Noah's Ark and the Holy Spirit⁴⁵—and the negative, mundane, and nuisance aspects to the "pigeon" (the common urban bird), human culture effectively "re-brands" the animal to maintain desired symbolic coherence. This allows for the simultaneous veneration of the "peace dove" while demonizing the "urban pigeon," revealing a human tendency to selectively interpret and categorize nature to fit evolving social narratives and practical realities, rather than confronting the inherent duality within the species itself. This process reflects a broader mechanism of cultural adaptation to complex or inconvenient truths.

Military propaganda, particularly during the World Wars, also contributed to the cultural representation of pigeons. Posters and official articles promoted their use, highlighting their reliability and importance to the war effort¹⁸, thereby shaping public perception of the "war pigeon" as a vital and heroic contributor.

5. Military Leadership Perspectives on Pigeon Warfare

The extensive and formalized integration of pigeons into military operations reflects a pragmatic and, at times, deeply appreciative perspective from military leaders and strategists. Their views were shaped by the pigeons' proven reliability and unique advantages in challenging combat environments.

Documented Statements from Generals and Military Leaders about Pigeon

Reliability

While direct, explicit quotes from top-tier generals like Patton or Eisenhower on pigeons are scarce in the provided material, the sheer scale of pigeon deployment and the consistent official reports of their effectiveness strongly imply a high regard from military leadership. The widespread adoption of pigeon services across the U.S., British, French, and German militaries in WWI and WWII underscores their perceived and actual value.¹

Pigeons were frequently lauded for their reliability, achieving a remarkable "95 percent success rate" in message delivery during WWI.¹ Official reports from the U.S. Naval Air Station Brest, France, highlighted a bird that delivered "97 messages during the last month of the war from sea planes at sea & in one y[ea]r covered a total distance of 1,638 miles delivering messages that saved many lives & naval aircraft".¹ Major Charles Whittlesey's critical reliance on his last pigeon, Cher Ami, to save the "Lost Battalion" in WWI, a mission that ultimately saved over 200 American lives, exemplifies the profound trust placed in these birds in desperate situations.⁶ Such instances underscore that pigeons were not merely a primitive fallback but a truly effective and often indispensable communication asset.

Official Military Policies and Protocols Regarding Pigeon Use

The institutionalization of military pigeon services demonstrates a formal, high-level strategic integration of these animals into national defense structures. The U.S. Army Signal Corps, for example, officially authorized a unit dedicated to training and managing homing pigeons in 1917, ultimately deploying over 50,000 pigeons for WWII alone.¹ This commitment extended to establishing specialized infrastructure, such as mobile lofts crafted from horse-drawn carriages or double-decker buses, enabling pigeons to operate directly on the front lines.²²

Military protocol dictated meticulous training regimens, including specific feeding schedules, gradual increases in flight distances for young birds, and daily exercise to maintain peak performance.¹⁴ Messages were handled with precision, often sent in duplicate using metal holders, and birds were released under specific conditions, avoiding night or heavy fog.¹⁶ The significant human effort and ingenuity invested in optimizing animal performance for military ends, from training to specialized

equipment, highlights a complex human-animal military partnership.

Furthermore, the strategic importance of pigeons was codified in law. Nations like the U.S. and the UK enacted legislation to protect military pigeons, making their destruction a serious offense punishable by imprisonment or fines.¹⁸ The U.S. Army even maintained a dedicated Veterinary Service unit focused on pigeon health, housing, and transport, underscoring the state's investment in and reliance on these biological assets.⁴³ These comprehensive policies and legal protections signify that pigeons were regarded as critical military resources, akin to human soldiers, rather than mere expendable tools.

Medals and Honors Awarded to Pigeons

The recognition of pigeons' contributions extended beyond mere utility to formal honors, mirroring those bestowed upon human combatants. Numerous pigeons received prestigious military decorations for their bravery and life-saving missions. Cher Ami, the famous WWI pigeon, was awarded the French Croix de Guerre with Palm for her heroic service.⁴ Similarly, the British Dickin Medal, instituted in 1943 by Maria Dickin, specifically honors "conspicuous gallantry or devotion to duty displayed by animals serving with the Armed Forces or Civil Defence units".⁴⁰ Often referred to as "the animals' Victoria Cross"⁴⁰, this medal has been awarded to 32 pigeons, celebrating their instrumental role in rescuing downed aircrews, delivering critical intelligence, and saving lives.¹⁸

Notable Dickin Medal recipients include White Vision, Winkie, and Tyke (the inaugural recipients in 1943 for aircrew rescues)³⁹, G.I. Joe (who saved a village in Italy from bombing)³⁷, and others like Beach Comber, Gustav, Paddy, Kenley Lass, and William of Orange for their vital wartime communications.⁴⁰ The act of awarding military medals to animals is a powerful act of anthropomorphization, attributing human-like qualities of bravery, devotion, and sacrifice to non-human actors. This practice served not only to acknowledge the pigeons' vital practical contributions but also to bolster human morale, create relatable narratives of heroism, and reinforce a sense of shared purpose and sacrifice across species in wartime. It reflects a military culture that, while pragmatic in its use of animals, also engaged in the emotional and symbolic integration of these animals into its ranks, constructing a narrative of "animal heroism" that mirrored and reinforced human ideals of valor.

Table 3: Notable Pigeon Recipients of the Dickin Medal

Pigeon Name	Date of Award	Heroic Act
White Vision	2 December 1943	Delivered message leading to rescue of ditched aircrew, flew 9 hours in bad visibility.
Winkie	2 December 1943	Delivered message contributing to rescue of ditched aircrew.
Tyke	2 December 1943	Delivered message contributing to rescue of ditched aircrew.
Beach Comber	1 September 1944	Brought first news of Dieppe landings in 1942.
Gustav	1 September 1944	Brought first message from Normandy beaches on D-Day (June 6, 1944).
Paddy	1 September 1944	Held best recorded time with a message from Normandy Operations.
Kenley Lass	March 1945	First pigeon to deliver intelligence from an agent in enemy-occupied France (October 1940), flew 300 miles in less than 7 hours.
G.I. Joe	(Awarded September 1946)	Saved lives in Calvi Vecchia, Italy, by delivering message that called off an air raid. ¹⁸

This table provides concrete examples of individual pigeons whose actions earned them the highest honors for animal bravery. It serves as a powerful testament to the individual contributions of these birds, grounding the abstract concept of "military utility" in tangible acts of heroism. This structured presentation reinforces the human tendency to recognize and celebrate non-human service in a manner akin to human valor, providing compelling examples for the report's narrative and directly addressing a key area of investigation.

6. The Transition from Military Asset to Peace Symbol

The historical trajectory of the pigeon, from a vital military asset to a pervasive peace emblem, represents a compelling case study in the fluidity of symbolic meaning. This transition was influenced by a complex interplay of technological advancements, evolving cultural perceptions, and deliberate artistic and political interventions.

Historical Points of Transition in Symbolic Meaning

The duality of the pigeon is not a recent phenomenon; the dove's symbolism as peace, purity, and the Holy Spirit predates its widespread military utility by millennia.⁴⁵ Early Christians, for instance, used the dove and olive branch to symbolize the peace of the soul and, later, civil peace, a tradition rooted in the Noah's Ark narrative.⁴⁹ Despite this ancient and enduring peaceful symbolism, pigeons were concurrently and extensively used in warfare across various civilizations for thousands of years.¹ This demonstrates a long-standing parallel existence of both symbolic meanings and practical, often violent, applications, suggesting human societies are capable of compartmentalizing these contradictory roles.

A significant point of transition in the popularization of the dove as a *secular* peace symbol occurred in the aftermath of World War II. This shift was largely catalyzed by Pablo Picasso's iconic "Dove of Peace" lithograph in 1949, which was adopted as the emblem for the World Peace Council and subsequently became a global symbol for various peace movements, including those advocating for nuclear disarmament.⁴⁹ This period marked a conscious and deliberate cultural re-branding of the bird, driven by the new anxieties of the nuclear age, where the need for universal peace became paramount.

Factors Influencing the Semantic Separation Between "Pigeons" and "Doves"

The linguistic and cultural distinction between "pigeon" and "dove" is a powerful

manifestation of how human perception shapes symbolic meaning. Scientifically, "pigeon" and "dove" are interchangeable terms referring to birds within the same family, Columbidae.⁵⁶ However, colloquially, "dove" typically denotes smaller, often white, or aesthetically pleasing birds, while "pigeon" commonly refers to the larger, often grey, urban bird.⁵⁸

This semantic divergence is not accidental; it is a cultural mechanism. The negative modern perception of urban pigeons as "flying rats," driven by concerns over disease transmission and the nuisance of droppings⁴⁵, contributes significantly to this deliberate separation. This cultural "cleansing" allows the powerful, ancient symbolism of peace and purity associated with the "dove" to remain untarnished by the practical realities and negative connotations of the common "pigeon." By culturally distinguishing "doves" from "pigeons," society effectively created a symbolic firewall, preserving an idealized representation of peace while externalizing the less desirable aspects of the same biological entity onto a different, more mundane label. This process is particularly pronounced in the post-WWII era, where the urgency of peace demanded a universally recognizable and unblemished symbol, highlighting a human tendency to selectively interpret and categorize nature to fit evolving social narratives and practical realities.

Persistence of the Dove as Peace Symbol in Post-Nuclear Age

The dove's association with peace, firmly established by early Christians, gained widespread secular recognition and intensified in the post-World War II nuclear age. It became a global symbol for various peace movements, including the Campaign for Nuclear Disarmament, reflecting the profound global anxiety surrounding atomic warfare.⁵⁰ This symbolic power endured and even grew, even as military pigeon services, which had been so vital during the World Wars, were systematically discontinued in the 1950s due to rapid advancements in electronic communication systems, radar, and other modern technologies.¹⁹ The obsolescence of their practical military role did not diminish their symbolic resonance; instead, the dove's image became an even more potent representation of humanity's aspiration for global harmony.

Contemporary Reconciliation of the Pigeon's Dual Symbolic Heritage

In recent times, there has been a discernible effort to reconcile the pigeon's dual symbolic heritage, acknowledging its past military service alongside its enduring status as a peace emblem. Practices such as the release of homing pigeons at Remembrance Day ceremonies ⁶⁸ and the unveiling of plaques commemorating their wartime contributions ⁶⁸ represent a growing recognition of the "war pigeon" legacy. Memorial artworks, such as ceramic doves inscribed with the names of World War I soldiers, explicitly bridge the gap between the bird's military service and its peaceful symbolism.⁶⁹

This process reflects a complex dialectic between historical utility and enduring symbolism in collective memory. As the direct military utility of pigeons faded, their symbolic role as messengers of peace became dominant, especially in the nuclear age where peace was paramount. However, historical memory, particularly in contexts like war memorials, now seeks to reclaim and honor their past sacrifices, even if that past utility (warfare) seems to contradict their primary symbolic association (peace). This suggests that societies actively curate and re-interpret the roles of non-human actors in their history, not just based on their functional purpose, but also on the evolving moral and cultural narratives they wish to uphold. The re-integration of the "war pigeon" into the "peace dove" narrative is an attempt to create a more complete, albeit complex, historical truth, acknowledging both the practical and symbolic dimensions of human-animal relations.

Conclusion: Reconciling Duality and Future Implications

This research has comprehensively explored the profound duality of pigeons in human history, revealing their simultaneous existence as indispensable instruments of war and powerful emblems of peace. From their ancient role as reliable battlefield messengers to Skinner's ambitious attempts at avian-guided missiles, pigeons consistently demonstrated unique biological and behavioral capabilities that often surpassed contemporary mechanical and electronic alternatives. Their innate homing instincts, superior visual acuity, remarkable trainability, and physical resilience made them invaluable assets for communication, reconnaissance, search and rescue, and even psychological operations. These attributes underscore a period in history where organic systems offered a robust and often superior solution to the limitations of

nascent human-engineered technologies.

The report has also highlighted the complex interplay between technological advancement and cultural perception. The eventual obsolescence of military pigeon services, driven by the rapid maturation of electronic communication and radar in the mid-20th century, coincided with the popularization of the dove as a universal peace symbol, particularly in the post-World War II nuclear age. This semantic divergence, where "dove" became associated with purity and peace while "pigeon" often devolved into a "flying rat" stereotype, illustrates how human societies construct and manipulate meaning to reconcile inherent contradictions and serve evolving social narratives. The military's rejection of technically viable animal-based systems, such as Project Pigeon, also demonstrated a cultural bias towards "less eccentric" mechanical solutions, even when animal performance was exceptional.

This study contributes significantly to the academic discourse by:

- Providing a holistic analysis of pigeons as both socio-technical actors and cultural symbols, thereby bridging disciplinary divides between Science and Technology Studies, military history, and cultural studies.
- Reframing our understanding of human-animal relations in warfare, moving beyond simplistic narratives to explore the strategic and symbolic integration of non-human agents within human technological and social systems.
- Offering new insights into the social and cultural factors that influence the adoption and rejection of unconventional technologies, demonstrating how perceptions of legitimacy and aesthetic preferences can override pragmatic technical efficacy.
- Developing a theoretical framework for understanding how a single species can embody profoundly contradictory cultural meanings, illustrating the dynamic and fluid nature of symbolism in human societies.

Ultimately, the pigeon's journey from battlefield hero to peace icon offers a unique lens through which to examine the dynamic relationship between technology, culture, and symbolism in human societies. It reminds us that the meaning and value we ascribe to the natural world are fluid, shaped by our needs, our fears, and our aspirations for both conflict and harmony. The ongoing efforts to reconcile the "war pigeon" with the "peace dove" in contemporary memory reflect a maturing historical consciousness, seeking to embrace the full, complex truth of our shared past, acknowledging the multifaceted roles that even the most unassuming creatures have played in shaping human destiny.

Works cited

1. The Human Relationship with Pigeons: Forgotten War Heroes | National Agricultural Library, accessed July 10, 2025, <https://www.nal.usda.gov/collections/stories/human-relationship-pigeons-forgotten-war-heroes>
2. How Do Homing Pigeons Find Home? - Wonderopolis, accessed July 10, 2025, <https://www.wonderopolis.org/wonder/how-do-homing-pigeons-find-home>
3. en.wikipedia.org, accessed July 10, 2025, https://en.wikipedia.org/wiki/War_pigeon#:~:text=6%20External%20links-.History,various%20parts%20of%20his%20empire.
4. How War Pigeons Changed the Course of Battle—and History, accessed July 10, 2025, <https://explorethearchive.com/war-pigeons>
5. Homing in Pigeons: The Role of the Hippocampal Formation in the Representation of Landmarks Used for Navigation - PMC - PubMed Central, accessed July 10, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6782352/>
6. Cher Ami - National Museum of the United States Army, accessed July 10, 2025, <https://www.thenmusa.org/biographies/cher-ami/>
7. Communications History: Homing Pigeons - Ooma, Inc., accessed July 10, 2025, <https://www.ooma.com/blog/communications-history-homing-pigeons/>
8. That Time the U.S. Coast Guard Had a Pigeon-Powered Sensor | by ..., accessed July 10, 2025, <https://medium.com/war-is-boring/that-time-the-u-s-coast-guard-had-a-pigeon-powered-sensor-147db4f4482b>
9. Bird vision - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Bird_vision
10. B F Skinner's Project Pigeon - YouTube, accessed July 10, 2025, https://www.youtube.com/watch?v=QgjUuW_gaBU
11. Pigeons, Operant Conditioning, and Social Control - Hack Education, accessed July 10, 2025, <http://hackeducation.com/2018/06/15/pigeons>
12. Project Pigeon - B. F. Skinner Foundation, accessed July 10, 2025, <https://www.bfskinner.org/project-pigeon/>
13. B.F. Skinner's Pigeon-Guided Rocket - Smithsonian Magazine, accessed July 10, 2025, <https://www.smithsonianmag.com/smithsonian-institution/bf-skinners-pigeon-guided-rocket-53443995/>
14. Pigeons of War - America in WWII magazine, accessed July 10, 2025, <http://www.americainwwii.com/articles/pigeons-of-war/>
15. Feathered Fighters: How the Carrier Pigeon Helped Win WWII - Uniformed Services Benefit Association, accessed July 10, 2025, <https://www.usba.com/blog/356/feathered-fighters-how-the-carrier-pigeon-helped-win-wwii>
16. Pigeons in War | Australian War Memorial, accessed July 10, 2025, <https://www.awm.gov.au/articles/encyclopedia/pigeons-in-war>
17. Carrier Pigeons in the First World War | Peace Palace Library, accessed July 10,

- 2025, <https://peacepalacelibrary.nl/blog/2014/carrier-pigeons-first-world-war>
18. War pigeon - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/War_pigeon
 19. America's Kaiser: How a pigeon served in two World Wars, accessed July 10, 2025, <https://www.thenmusa.org/americas-kaiser-how-a-pigeon-served-in-two-world-wars/>
 20. B.F. Skinner: The Man Who Taught Pigeons to Play Ping-Pong and Rats to Pull Levers, accessed July 10, 2025, <https://www.smithsonianmag.com/science-nature/bf-skinner-the-man-who-taught-pigeons-to-play-ping-pong-and-rats-to-pull-levers-5363946/>
 21. It's a bird! It's a plane! | Article | The United States Army, accessed July 10, 2025, https://www.army.mil/article/111511/its_a_bird_its_a_plane
 22. The Heroic WWI Homing Pigeons That Saved Countless Lives | HISTORY, accessed July 10, 2025, <https://www.history.com/articles/heroic-world-war-i-homing-pigeons-cher-ami>
 23. The Incredible Carrier Pigeons of the First World War | IWM, accessed July 10, 2025, <https://www.iwm.org.uk/history/the-incredible-carrier-pigeons-of-the-first-world-war>
 24. Project Pigeon, accessed July 10, 2025, https://www3.uca.edu/igzoo/Exhibits/project_pigeon.htm
 25. 1940 - Project Pigeon (1948 - Project Orcon) - B.F. Skinner (American) - cyberneticzoo.com, accessed July 10, 2025, <https://cyberneticzoo.com/bionics/1940-project-pigeon-1948-project-orcon-b-f-skinner-american/>
 26. Project Pigeon - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Project_Pigeon
 27. Project Pigeon/Project Orcon | Secret Projects Forum, accessed July 10, 2025, <https://www.secretprojects.co.uk/threads/project-pigeon-project-orcon.10744/>
 28. The Pigeon in the Machine. The Concept of Control in Behaviorism and Cybernetics, accessed July 10, 2025, <https://d-nb.info/1234470411/34>
 29. Norbert Wiener - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Norbert_Wiener
 30. Norbert and Gregory: Two strands of cybernetics - Open Research Online, accessed July 10, 2025, <https://oro.open.ac.uk/19485/2/E3A57887.pdf>
 31. Norbert Wiener Publishes the Book Cybernetics | aiws.net, accessed July 10, 2025, <https://aiws.net/the-history-of-ai/aiws-house/cybernetics-norbert-wiener-publishes-the-book-cybernetics/>
 32. B.F. Skinner's Nose Cone of a Pigeon-Guided Missile, accessed July 10, 2025, https://americanhistory.si.edu/ne/collections/object/nmah_690069
 33. en.wikipedia.org, accessed July 10, 2025, https://en.wikipedia.org/wiki/War_pigeon#:~:text=In%20Ancient%20Rome%2C%20Julius%20Caesar.hawks%20to%20hunt%20the%20pigeons.

34. The British Army entrusted its secrets to birdbrains, accessed July 10, 2025, <https://www.nam.ac.uk/explore/british-army-entrusted-its-secrets-birdbrains>
35. Unsung heroes of World War I: the carrier pigeons - Pieces of History, accessed July 10, 2025, <https://prologue.blogs.archives.gov/2018/01/08/unsung-heroes-of-world-war-i-the-carrier-pigeons/>
36. Cher Ami - Home of Heroes, accessed July 10, 2025, <https://homeofheroes.com/heroes-stories/world-war-i/cher-ami/>
37. News - Army Sells Its Last Homing Pigeon - DVIDS, accessed July 10, 2025, <https://www.dvidshub.net/news/440930/army-sells-its-last-homing-pigeon>
38. Pigeons at War - The RAF and the National Pigeon Service - Classic Warbirds, accessed July 10, 2025, <https://www.classicwarbirds.co.uk/articles/pigeons-at-war-the-raf-and-the-national-pigeon-service.php>
39. Three pigeons were awarded a military medal in 1943. - History Facts, accessed July 10, 2025, <https://historyfacts.com/world-history/fact/three-pigeons-were-awarded-a-military-medal-in-1943/>
40. Dickin Medal - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Dickin_Medal
41. Source Columba - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Source_Columba
42. Pigeons in Parachutes Carried Messages During World War II - My Modern Met, accessed July 10, 2025, <https://mymodernmet.com/pigeon-parachutes-world-war-ii/>
43. United States Army Pigeon Service - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/United_States_Army_Pigeon_Service
44. Military Carrier Pigeons in the Era of Electronic Warfare - Schneier on Security, accessed July 10, 2025, https://www.schneier.com/blog/archives/2019/01/military_carrie.html
45. Pigeon Symbolism: Unraveling the Meanings and Myths | Birdfact, accessed July 10, 2025, <https://birdfact.com/articles/what-do-pigeons-symbolize>
46. Pigeon; the reflection of purity and peace, accessed July 10, 2025, <https://www.aensiweb.com/old/jasa/rjfh/2013/110-113.pdf>
47. Dove - symbol of peace and the Holy Spirit, accessed July 10, 2025, <https://lcumc.org/devotionals/f/dove-%E2%80%93-symbol-of-peace-and-the-holy-spirit>
48. The Enduring Symbolism of Doves - Biblical Archaeology Society, accessed July 10, 2025, <https://www.biblicalarchaeology.org/daily/ancient-cultures/daily-life-and-practice/the-enduring-symbolism-of-doves/>
49. Doves as symbols - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Doves_as_symbols
50. Peace symbols - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Peace_symbols

51. The Dove As A Political Symbol For Peace - FasterCapital, accessed July 10, 2025, <https://fastercapital.com/topics/the-dove-as-a-political-symbol-for-peace.html/1>
52. Peace Sign and Enduring Symbol at Folk Festivals, accessed July 10, 2025, <https://mariposafolk.com/peace-sign-enduring-symbol-folk-festivals/>
53. en.wikipedia.org, accessed July 10, 2025, https://en.wikipedia.org/wiki/Peace_symbols#:~:text=The%20dove%20symbol%20was%20used,metamorphosing%20into%20a%20Soviet%20tank.
54. Brave Pigeons of Wartime | The British Newspaper Archive Blog, accessed July 10, 2025, <https://blog.britishnewspaperarchive.co.uk/2020/07/03/brave-pigeons-of-wartime/>
55. PDSA Dickin Medal, accessed July 10, 2025, <https://www.pdsa.org.uk/what-we-do/animal-awards-programme/pdsa-dickin-medal>
56. A Symbol | Hope Lutheran Church, accessed July 10, 2025, <https://www.hopewarren.com/2021/10/a-symbol/>
57. Pigeon post - Wikipedia, accessed July 10, 2025, https://en.wikipedia.org/wiki/Pigeon_post
58. Doves & Pigeons - Nature in Novato, accessed July 10, 2025, <https://natureinnovato.com/2018/09/26/doves-and-pigeons/>
59. Dove vs Pigeon - Reddit, accessed July 10, 2025, https://www.reddit.com/r/pigeon/comments/1dlo3bk/dove_vs_pigeon/
60. What's the Difference Between Pigeons and Doves? - Mental Floss, accessed July 10, 2025, <https://www.mentalfloss.com/article/554182/what-is-difference-between-pigeons-and-doves>
61. I just signed a Christmas card at work calling the dove a 'universal symbol of peace'. When and why/how did the dove become a symbol of peace? : r/AskHistorians - Reddit, accessed July 10, 2025, https://www.reddit.com/r/AskHistorians/comments/1st0am/i_just_signed_a_christmas_card_at_work_calling/
62. A Hero Pigeon of World War I | National Archives, accessed July 10, 2025, <https://www.archives.gov/publications/prologue/2009/winter/hero-pigeon-president-wilson>
63. THE MILITARY USE OF THE HOMING PIGEON, accessed July 10, 2025, <https://sora.unm.edu/sites/default/files/journals/wilson/v039n02/p0067-p0074.pdf>
64. The Legal History of Pigeons | In Custodia Legis - Library of Congress Blogs, accessed July 10, 2025, <https://blogs.loc.gov/law/2021/09/the-legal-history-of-pigeons/>
65. CECOM HISTORY IS FOR THE BIRDS: HERO PIGEONS | Article - Army.mil, accessed July 10, 2025, https://www.army.mil/article/74924/cecom_history_is_for_the_birds_hero_pigeons
66. Carrier Pigeon Service - The Long, Long Trail, accessed July 10, 2025, <https://www.longlongtrail.co.uk/army/regiments-and-corps/the-corps-of-royal-engineers-in-the-first-world-war/carrier-pigeon-service/>

67. Top Tips For Managing Your Carrier Pigeons | Imperial War Museums, accessed July 10, 2025,
<https://www.iwm.org.uk/history/top-tips-for-managing-your-carrier-pigeons>
68. Essential Allies: The History of the Wartime Homing Pigeon - RSL Victoria, accessed July 10, 2025,
<https://rslvic.com.au/news/essential-allies-the-history-of-the-wartime-homing-pigeon/>
69. The meaning behind the doves - Ipswich School, accessed July 10, 2025,
<https://www.ipswich.school/meaning-behind-the-doves/>